Answer Key & Teacher Notes

Session 1 — Polygons and Perimeter on the Plane · Session 2 — Designing Polygons with Scale

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Polygons and Perimeter on the Plane

TALK PROMPTS

- How do matching coordinates tell you which sides run across and which run up and down?
- Why does finding only two side lengths give you the whole perimeter of a rectangle?

PARTNER TASK

Garden Plot (Form A, problem 2). The plot has corners (−4, 6), (9, 6), (9, −3) and (−4, −3). One partner finds the top side; the other finds the left side. Together find the perimeter (13 + 9 + 13 + 9 = 44 m) and explain why two side lengths give the whole perimeter.

WATCH FOR

- Subtracting coordinates with opposite signs ((−5, 4) to (3, 4) as 2 instead of 8).
- Adding only two sides of a rectangle for the perimeter.
- Mixing up perimeter (units) and area (square units), or forgetting the 1/2 in triangle area.

SESSION 2 · TEACHER NOTES

Designing Polygons with Scale

TALK PROMPTS

- How does the scale change your answer even though the drawing stays the same size?
- Why should you change each side to real length before you find the area?

PARTNER TASK

Fence Cost (Form A, problem 6). The yard has corners (−4, 3), (8, 3), (8, −5) and (−4, −5); each unit is 4 feet. One partner finds the side lengths in grid units; the other changes them to feet. Together find the perimeter (160 ft) and check it by multiplying the grid perimeter of 40 units by 4.

WATCH FOR

- Leaving a side in grid units (12 units written as 12 ft).
- Multiplying the grid area by the scale only once instead of changing both sides to real length.
- Multiplying instead of dividing to find a missing side from an area.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.3	1	2	1	2	2	2	2	3	2	1	1
S1 · Form B	6.G.A.3	1	2	1	2	2	2	2	3	2	1	1
S2 · Form A	6.G.A.3	1	2	2	1	1	2	2	3	2	1	1
S2 · Form B	6.G.A.3	1	2	2	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

polygon: A closed shape with **straight** sides.

vertex: A **corner** point of a polygon.

perimeter: The distance all the way **around** a shape.

area: The number of **square** units inside a shape.

- Two vertices that share a coordinate make a **horizontal** or vertical side.
- Perimeter is the **sum** of all the sides. Area of a rectangle is length **×** width.

Example: Top side: same y, opposite signs, so 1 + 5 = **6** units. Right side: same x, opposite signs, so 4 + 3 = **7** units. Perimeter: 6 + 7 + 6 + 7 = **26** units. Answer: Area = 6 × 7 = **42** square units

YOU TRY

1. Each side is 3 + 1 = 4 units; P = 4 × 4 = 16 units

2. (−4, −5)

EXIT TICKET

1. a. 6 + 3 = 9 units b. 2 + 4 = 6 units

2. P = 9 + 6 + 9 + 6 = 30 units; A = 9 × 6 = 54 square units

3. **A** (adding all the sides)

PRACTICE FORM A

1. a. 3 + 6 = 9 units b. 4 + 2 = 6 units c. 9 + 6 + 9 + 6 = 30 units d. 9 × 6 = 54 square units

2. a. rectangle plotted b. 4 + 9 = 13 m c. 6 + 3 = 9 m d. 13 + 9 + 13 + 9 = 44 m

3. 7 + 11 = 18 m and 10 + 1 = 11 m; P = 18 + 11 + 18 + 11 = 58 m

4. (−2, 1)

5. base 5 + 7 = 12, height 2 + 5 = 7; 1/2 × 12 × 7 = 42 square units

6. 6 + 7 = 13 yd and 9 − 2 = 7 yd; 13 × 7 = 91 square yards

7. 12 + 6 + 5 + 6 + 7 + 12 = 48 ft

8. a. 15 units by 9 units b. 48 units; 48 × 7 = $336 c. 135 square units; 135 × 2 = $270

9. The x-coordinates −5 and 3 have opposite signs, so the side crosses the y-axis. Jorge should add: 5 + 3 = 8 units.

10. **B** (26 units)

11. **C** (40 square units)

Self-check word **VERTEX**

PRACTICE FORM B

1. a. 5 + 5 = 10 units b. 3 + 4 = 7 units c. 10 + 7 + 10 + 7 = 34 units d. 10 × 7 = 70 square units

2. a. rectangle plotted b. 5 + 7 = 12 m c. 5 + 4 = 9 m d. 12 + 9 + 12 + 9 = 42 m

3. 6 + 10 = 16 m and 9 + 3 = 12 m; P = 16 + 12 + 16 + 12 = 56 m

4. (−3, 2)

5. base 4 + 6 = 10, height 3 + 5 = 8; 1/2 × 10 × 8 = 40 square units

6. 5 + 8 = 13 yd and 7 − 1 = 6 yd; 13 × 6 = 78 square yards

7. 11 + 6 + 4 + 6 + 7 + 12 = 46 ft

8. a. 14 units by 9 units b. 46 units; 46 × 6 = $276 c. 126 square units; 126 × 3 = $378

9. The x-coordinates −9 and 6 have opposite signs, so the side crosses the y-axis. Bri should add: 9 + 6 = 15 units.

10. **B** (26 units)

11. **C** (36 square units)

Self-check word **CORNER**

SESSION 2 · ANSWERS

Designing Polygons with Scale

GUIDED NOTES

scale: How much real length one grid **unit** stands for.

area: The flat space **inside** a shape, in square units.

dimensions: The **length** and width of a shape.

- Multiply grid units by the **scale** to get real length.
- To find a missing side, **divide** the area by the known side.

Example: Real width: 5 × 4 = **20** m. Real height: 3 × 4 = **12** m. Answer: Area = 20 × 12 = **240** m²

YOU TRY

1. a. 8 × 5 = 40 ft b. 35 ÷ 5 = 7 units

2. 600 ÷ 20 = 30 ft

EXIT TICKET

1. a. 27 m b. 16 units

2. 20 ft × 30 ft = 600 ft²

3. **B** (divide)

PRACTICE FORM A

1. a. 24 ft b. 54 ft c. 7 units d. 15 units

2. a. (−5, −3) b. 11 units = 110 ft; 9 units = 90 ft c. 110 × 90 = 9,900 ft²

3. 12 units = 36 ft; 9 units = 27 ft; 36 × 27 = 972 ft²

4. 150 ft and 100 ft; 150 + 100 + 150 + 100 = 500 ft

5. 30,000 ÷ 150 = 200 ft

6. 12 units = 48 ft; 8 units = 32 ft; P = 160 ft; 160 × 6 = $960

7. 12 units = 24 ft; 384 ÷ 24 = 16 ft

8. a. 8 × 15 = 120 ft b. 5,400 ÷ 120 = 45 ft c. 45 ÷ 15 = 3 units d. 120 + 45 + 120 + 45 = 330 ft

9. 12 is the length in grid units (3 + 9). Talia forgot the scale. Multiply: 12 × 6 = 72 feet.

10. **A** (30 ft)

11. **C** (100 feet)

Self-check word **DESIGN**

PRACTICE FORM B

1. a. 40 ft b. 56 ft c. 6 units d. 12 units

2. a. (−6, −4) b. 12 units = 120 ft; 9 units = 90 ft c. 120 × 90 = 10,800 ft²

3. 12 units = 24 ft; 9 units = 18 ft; 24 × 18 = 432 ft²

4. 100 ft and 60 ft; 100 + 60 + 100 + 60 = 320 ft

5. 4,200 ÷ 70 = 60 ft

6. 12 units = 36 ft; 7 units = 21 ft; P = 114 ft; 114 × 8 = $912

7. 9 units = 18 ft; 360 ÷ 18 = 20 ft

8. a. 6 × 10 = 60 ft b. 4,800 ÷ 60 = 80 ft c. 80 ÷ 10 = 8 units d. 60 + 80 + 60 + 80 = 280 ft

9. 12 is the length in grid units (7 + 5). Kenji forgot the scale. Multiply: 12 × 8 = 96 feet.

10. **B** (30 m)

11. **C** (90 meters)

Self-check word **LAYOUT**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · POLYGONS AND PERIMETER ON THE PLANE

1. A(−4, 3), B(2, 3), C(2, −2), D(−4, −2). a. 4 + 2 = 6 units b. 3 + 2 = 5 units c. 6 + 5 + 6 + 5 = 22 units d. 6 × 5 = 30 square units

2. a. 3 + 4 = 7 units b. 2 + 4 = 6 units c. ½ × 7 × 6 = 21 square units

3. Sides 5 + 6 = 11 m and 3 + 4 = 7 m; P = 11 + 7 + 11 + 7 = 36 m

4. Each side is 3 + 2 = 5 units; P = 4 × 5 = 20 units

5. (−2, −1). It must line up with (−2, 5) vertically (x = −2) and with (3, −1) horizontally (y = −1).

SHEET 2 · DESIGNING POLYGONS WITH SCALE

1. A(−3, 4), B(−3, −2). a. 4 + 2 = 6 units b. 6 × 5 = 30 ft c. 1,200 ÷ 30 = 40 ft d. 40 ÷ 5 = 8 units

2. a. 28 m b. 9 units

3. 14 ft × 10 ft = 140 ft²

4. Grid sides 2 + 2 = 4 units (12 ft) and 4 + 1 = 5 units (15 ft). P = 12 + 15 + 12 + 15 = 54 ft

5. Lily subtracted 5 − 4. The signs differ, so add: 4 + 5 = 9 units; 9 × 3 = 27 ft.

6. **B** (30 m)

7. **A** (20 ft)

October 2026 conceptual additions — teacher key

Session 1 · Polygons and Perimeter on the Plane

Width 6, height 4; perimeter 20 units; area 24 square units. Missing corner (−2,−1) completes matching horizontal and vertical sides.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Width 6, height 4; perimeter 20 units; area 24 square units. Missing corner (−2,−1) completes matching horizontal and vertical sides.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Designing Polygons with Scale

Real dimensions 20×12 m; perimeter 64 m; area 240 m². Both dimensions scale by 4, so area scales by 16: 15×16=240.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Real dimensions 20×12 m; perimeter 64 m; area 240 m². Both dimensions scale by 4, so area scales by 16: 15×16=240.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.